

Python clicks when you build it.

Adaptive practice, immediate feedback, and an instructor-ready view of learning.

FDE + AI RESEARCH ENGINEER

DEPLOYED MVP

```
def is_palindrome(text):  
    clean = text.lower()  
    return clean == clean[::-1]  
  
print(is_palindrome(input()))
```

✓ All tests passed

One learning loop. Two clear surfaces.

Students get immediate coaching and mastery updates; instructors get cohort signal without digging through raw submission noise.

01

Practice

Open a focused Python assignment, write code in Monaco, and grade it in place.

02

Reflect

See score, feedback, mistakes, hint, mastery change, and the recommended next step.

03

Review

Instructors see one row per student, then drill into mastery and submission history.

AUTH

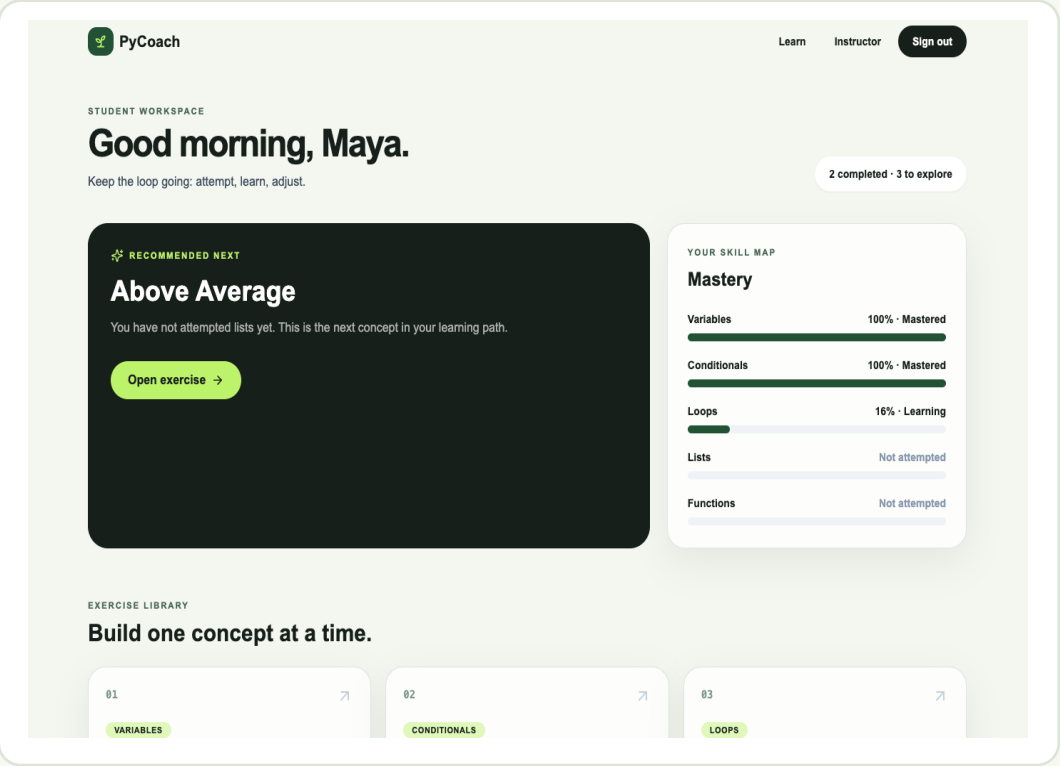
Email/password login, student self-signup, and clearly labeled demo accounts

RECOMMENDATION

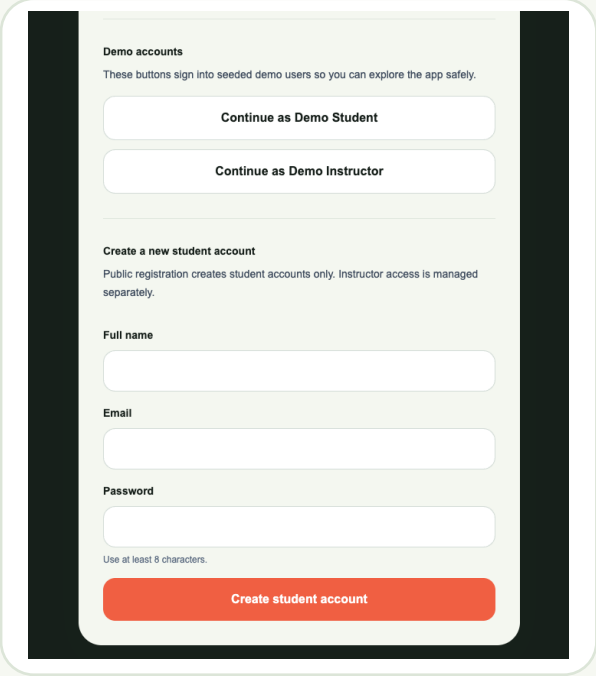
Start here → next unattempted skill → weakest mastery after evidence exists

The current product surface

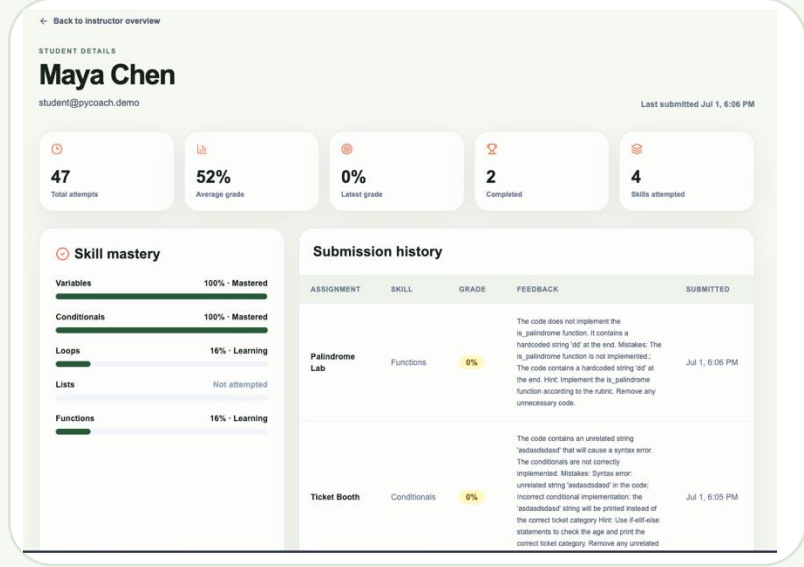
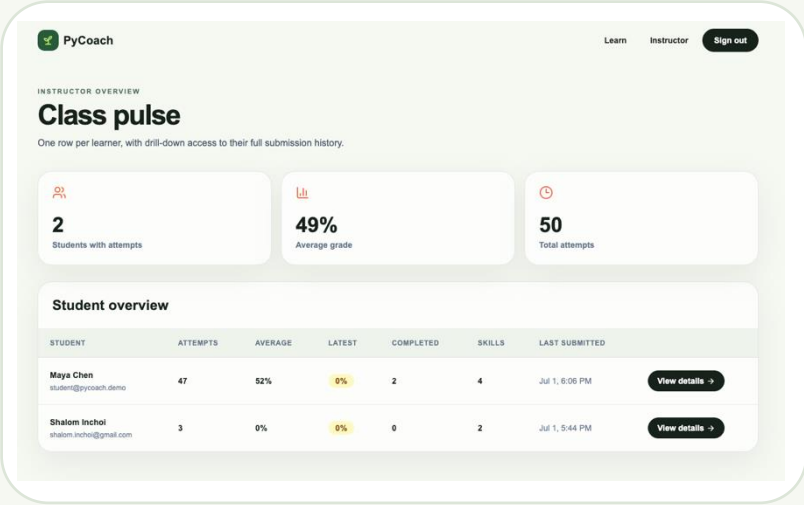
Screens below come from the deployed app and reflect the latest login, student, and instructor flows.



STUDENT DASHBOARD



LOGIN + SIGNUP



INSTRUCTOR DETAIL

[← Back to dashboard](#)

BEGINNER · VARIABLES

The Tip Splitter

Read a bill total and a positive number of people, then print each person's equal share rounded to exactly two decimal places.

Example tests

in 100 ↵ 4
out 25.00

in 99 ↵ 3
out 33.00

solution.py

▶ Grade solution

```
1 bill = float(input())
2 people = int(input())
3
4 # Calculate and print each person's share
5
```

Assignment

BEGINNER · VARIABLES

The Tip Splitter

Read a bill total and a positive number of people, then print each person's equal share rounded to exactly two decimal places.

Example tests

in 100 ↵ 4
out 25.00

in 99 ↵ 3
out 33.00

solution.py

▶ Grade solution

```
1 bill = float(input())
2 people = int(input())
3 print(f"{bill / people:.2f}")
```

Passed

100%

Your code correctly reads inputs, divides the bill by the number of people, and prints the share formatted to exactly two decimal places.

HINT

Make sure to handle cases where the bill total is zero or a decimal number without hardcoding them.

The code correctly implements the required functionality and handles edge cases.

UPDATED VARIABLES MASTERY

100%

RECOMMENDED NEXT

Above Average

You have not attempted lists yet. This is the next concept in your learning path.

[Continue to Recommended Assignment →](#)

[View dashboard](#)

PASSED

INTERMEDIATE · FUNCTIONS

Palindrome Lab

Implement `is_palindrome(text)`, then read one line and print True or False. Ignore letter case, spaces, and punctuation; an empty string is a palindrome.

Example tests

in Racecar
out True

in hello
out False

```
1 def is_palindrome(text):
2     # Return whether text reads the same backwards
3     pass
4
5 text = input()
6 print(is_palindrome(text))
7 ddd
```

Keep practicing

0%

The code does not implement the `is_palindrome` function. It contains a hardcoded string 'dd' at the end.

WHAT TO IMPROVE

- The `is_palindrome` function is not implemented.
- The code contains a hardcoded string 'dd' at the end.

HINT

Implement the `is_palindrome` function according to the rubric. Remove any unnecessary code.

The code does not solve the problem.

UPDATED FUNCTIONS MASTERY

16%

RECOMMENDED NEXT

Palindrome Lab

Review this skill and try the assignment again before moving on.

[Try Again](#)

[View dashboard](#)

Grader: Grog LLM

FAILED

[← Back to dashboard](#)

BEGINNER · VARIABLES

The Tip Splitter

Read a bill total and a positive number of people, then print each person's equal share rounded to exactly two decimal places.

Example tests

in 100 ↵ 4
out 25.00

in 99 ↵ 3
out 33.00

solution.py

▶ Grade solution

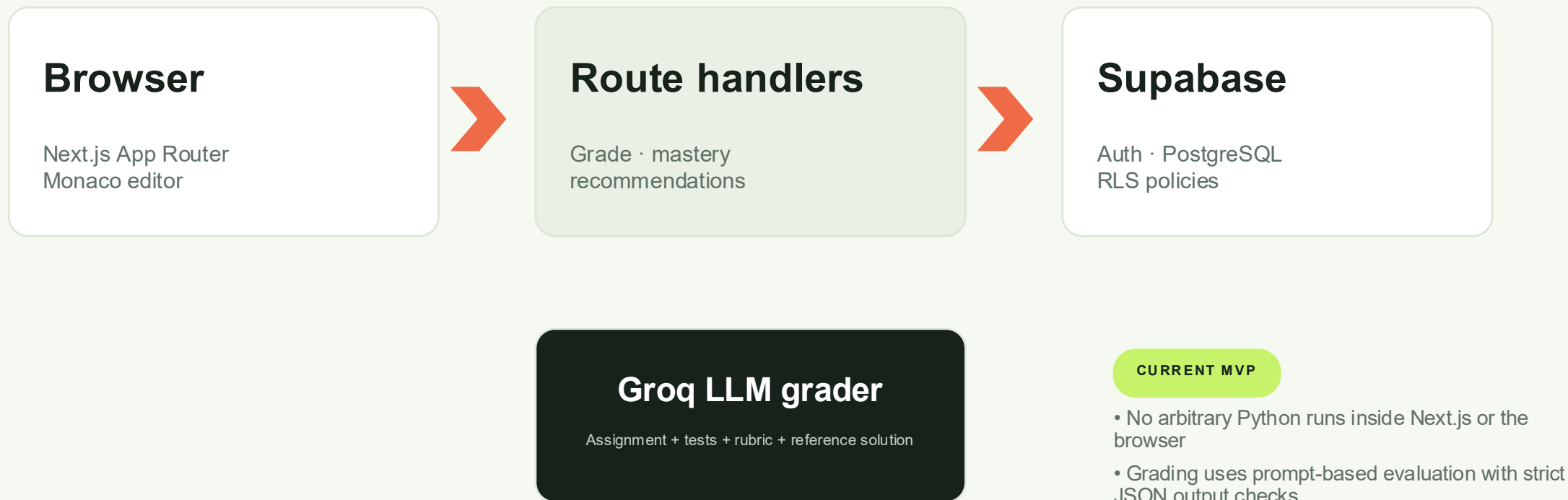
```
1 bill = float(input())
2 people = int(input())
3
4 # Calculate and print each person's share
5 bill = float(input())
6 people = int(input())
7 print(f"{bill / people:.2f}")
```

Syntax error: missing closing ")" for "(" on line 7.

SYNTAX ERROR

A deliberately small MVP surface

One Next.js codebase, Supabase for auth and persistence, and Groq for fast grading feedback.



Prompted for correctness, completeness, and useful feedback

Current MVP uses Groq with richer grading context; deterministic execution is the production upgrade.



← Back to dashboard

BEGINNER · VARIABLES

The Tip Splitter

Read a bill total and a positive number of people, then print each person's equal share rounded to exactly two decimal places.

Example tests

```
in 100 ↵ 4
out 25.00
```

```
in 99 ↵ 3
out 33.00
```

```
1 bill = float(input())
2 people = int(input())
3
4 # Calculate and print each person's share
5
```

Grade solution

- Prompt includes title, description, skill, starter code, public tests, hidden tests, rubric, and backend-only reference solution
- The grader is asked to check every required branch or case, not just the easiest visible example
- If syntax issues and missing logic both exist, feedback should mention both
- Output must be valid JSON before any grade is saved

PRODUCTION UPGRADE

Deterministic sandbox execution plus LLM explanation remains future work, not a current claim.

BKT is simple, interpretable, and honest about sparse data

The UI hides the 0.30 prior until a skill has evidence, then recommends what to do next based on observed work.

BEGINNER - VARIABLES

The Tip Splitter

Read a bill total and a positive number of people, then print each person's equal share rounded to exactly two decimal places.

Example tests

in: 100, 4
out: 25.00

in: 99, 3
out: 33.00

Passed

100%

Your code correctly reads inputs, divides the bill by the number of people, and prints the share formatted to exactly two decimal places.

HINT

Make sure to handle cases where the bill total is zero or a decimal number without hardcoding them.

The code correctly implements the required functionality and handles edge cases.

UPDATED VARIABLES MASTERY

100%

RECOMMENDED NEXT

Above Average

You have not attempted this yet. This is the next concept in your learning path.

Continue to Recommended Assignment

View dashboard

INTERMEDIATE - FUNCTIONS

Palindrome Lab

Implement is_palindrome(text), then read one line and print True or False. Ignore letter case, spaces, and punctuation; an empty string is a palindrome.

Example tests

in: Racecar
out: True

in: hello
out: False

Keep practicing

0%

The code does not implement the is_palindrome function. It contains a hardcoded string 'lol' at the end.

WHAT TO IMPROVE

The is_palindrome function is not implemented.

The code contains a hardcoded string 'lol' at the end.

HINT

Implement the is_palindrome function according to the rubric. Remove any unnecessary code.

The code does not solve the problem.

UPDATED FUNCTIONS MASTERY

16%

RECOMMENDED NEXT

Palindrome Lab

Review this skill and try the assignment again before moving on.

Try Again

View dashboard

$P(L_0)$

0.30

$P(T)$

0.12

$P(G)$

0.20

$P(S)$

0.10

Recommendation behavior

- Brand new student: start with Variables
- Partially attempted path: recommend the next unattempted concept
- All skills attempted: recommend the weakest mastery
- Score below 80 after grading: retry the same assignment first

PYCOACH / SHALOM INCHOI

03 JUL 2026 7

What is real today, and what comes next

The app is deployed, tested through build, and documented with clear next steps for production hardening.

LIVE

VERCEL APP

AUTH

LOGIN + SIGNUP

GRADE

GROQ + RUBRIC

OVERVIEW

INSTRUCTOR DRILL-DOWN

Research basis

- Corbett & Anderson (1995) · Knowledge Tracing: Modeling the Acquisition of Procedural Knowledge
<https://act-r.psy.cmu.edu/wordpress/wp-content/uploads/2012/12/893CorbettAnderson1995.pdf>
- Pardos & Heffernan (2010) · Modeling Individualization in a Bayesian Networks Implementation of Knowledge Tracing.
https://web.cs.wpi.edu/~nth/pubs_and_grants/papers/2010/PardosUser%20Modeling2010.pdf
- Piech et al. (2015) · Deep Knowledge Tracing
<https://arxiv.org/abs/1506.05908>

NEXT

Add deterministic sandbox execution, per-test feedback, and calibrated BKT from real learner sequences.